

SL No.X/Sem4 /I

Model Question Paper I

DC MACHINES & TRANSFORMERS

Time: 3 Hours

Max.Marks: 75

PART A

- I. Answer **all** questions in one word or one sentence. Each question carries **1 mark**.

1	Fleming's Right hand rule is applicable to -----	M1.01	R
2	In a DC Generator which part convert ac into dc	M1.01	R
3	How many parallel path are there in wave winding	M1.03	R
4	Most suitable motor for electric traction is	M2.01	R
5	If the DC series motor is started at no load it will run at -----	M2.03	R
6	Which winding in a transformer has more number of turns either high voltage or low voltage	M3.01	R
7	To step 110V AC down to 20V AC the turns ratio of transformer will be....	M3.02	R

8	The core in the large transformer is built of -----	M4.01	R
9	The saving in copper achieved by converting ordinary two winding transformer into an auto-transformer is governed by the.....	M4.04	R

PART B

- II. Answer any **eight** questions from the following, each question carries **3 marks**.
(8 * 3=24 marks)

1	Derive the expression for generated emf in DC generator.	M1.01	A
2	Explain significance of back emf	M1.01	U
3	What is an armature reaction?	M1.02	U
4	With the help of speed-armature current characteristics, explain why the series motors should not be started without any load.	M2.03	U
5	Define the process of commutation with neat sketches	M2.04	U
6	Draw the electrical characteristics of DC shunt motor	M2.01	R

7	Derive the condition for maximum efficiency of a single-phase transformer	M3.01	U
8	Define all day efficiency. How this efficiency of a transformer varies with load	M3.02	R
9	What are the different connection of three phase transformer	M4.03	R
10	What are the disadvantages of delta-delta connection	M4.04	R

PART C

Answer **ALL** questions. Each question carries **7 marks**.

(6*7=42 marks)

III	Explain working principle of DC machine and significance of back emf	M1.04	R
	OR		
IV	Explain the different methods of excitation of dc generator	M1.04	A
V	What is the necessity of a starter for a motor? With a suitable diagram, explain the working of 3 point starter	M2.03	R
	OR		
VI	Explain the speed control of separately excited dc motors	M2.01	R
VII	Derive an expression for the saving of copper in an autotransformer as compared to an equivalent two winding transformer	M2.04	U

	OR		
VIII	Draw the circuit diagram of Sumpner's test and derive the equation for efficiency of each transformer	M2.02	U
IX	Explain how the Scott connection can be used to obtain two phase supply from a three phase supply?	M3.04	U
	OR		
X	Explain the working of practical transformer with help of neat diagram	M3.03	U
XI	Develop the equivalent circuit of a transformer.	M3.01	A
	OR		
XII	.Explain the working of ideal transformer with help of neat diagram	M3.01	A

XIII	Explain the working of an instrument transformer with help of neat diagram	M4.02	A
	OR		
XIV	. A single-phase transformer has 1000 turns on the primary and 200 turns on the secondary. The no-load current drawn is 3 A at a power factor 0.2 lagging. Calculate the total primary current and power factor at primary when the secondary is loaded to a current of 280 A at a power factor of 0.8 lagging. Assume the voltage drop in the windings to be negligible.	M4.04	A

Module Wise Question Analysis

Module	Hr/ module	(hi / ΣHi) * 123	Type of questions							
			Part A (1 mark)		Part B (3mark)		Part C (7mark)		Total	
			No	Marks	No	Marks	No	Marks	No	Marks
I	15	28.60	3		3		2		8	
			-----	3	-----	9	-----	14	-----	26
II	14	34.32	2		3		4		9	
			-----	2	-----	9	-----	28	-----	39
III	14	28.60	2		2		4			
			-----	2	-----	6	-----	28	-----	36
IV	15	31.46	2		2		2		6	
			-----	2	-----	6	-----	14	-----	22
Total			9		10		12		31	
			-----	9	-----	30	-----	84	-----	123

Cognitive Level Wise Question Analysis

level	% mark distribu tion	mark s/leve l	Type of questions							
			Part A (1 mark)		Part B (3mark)		Part C (7mark)		Total	
			No	Marks	No	Marks	No	Marks	No	Marks
R	30	36.9	9		4		3		16	
			-----	9	-----	12	-----	21	-----	42
U	50	61.5	0		5		4		9	
			-----	0	-----	15	-----	28	-----	43
A	20	24.6	0		1		5		6	
			-----	0	-----	3	-----	35	-----	38
Total			9		10		12		31	
			-----	9	-----	30	-----	84	-----	123

Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.01	R	1	2
I.2	M1.02	R	1	2
I.3	M1.03	R	1	2
I.4	M2.01	R	1	2
I.5	M2.03	R	1	2
I.6	M3.01	R	1	2
I.7	M3.02	R	1	2
I.8	M4.01	R	1	2
I.9	M4.04	R	1	2
II.1	M1.01	A	3	7
II.2	M1.02	U	3	7
II.3	M1.02	U	3	8
II.4	M2.03	U	3	8
II.5	M2.04	U	3	7
II.6	M2.01	R	3	8
II.7	M3.01	U	3	7
II.8	M3.02	R	3	7
II.9	M4.02	R	3	7
II.10	M4.04	R	3	7
III.	M1.04	R	7	17
IV.	M1.04	A	7	17
V	M2.03	R	7	17
VI	M2.01	R	7	17
VII	M2.04	U	7	17

VIII	M2.02	U	7	17
IX	M3.02	U	7	17
X	M3.03	U	7	17
XI	M3.01	A	7	17
XII	M3.01	A	7	17
XIII	M4.02	A	7	17
XIV	M4.04	A	7	17
Total			123	295

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SL No.X/Sem3/II

Model Question Paper II

DC MACHINES & TRANSFORMERS

Time: 3 Hours

Max.Marks: 75

PART A

III. Answer **all** questions in one word or one sentence. Each question carries **1 mark**.

1	Which one of the materials are used for brushes in dc machine	M1.01	R
2	The series field of short shunt dc generator is excited by	M1.02	R
3	The current drawn by the armature of DC motor is directly proportional to -----	M2.01	R

4	Why is the field winding of a DC series motor provided with thick wire?	M2.02	R
5	An ideal transformer is one which has -----	M3.01	R
6	The transformer rating are usually expressed in terms of -----	M3.02	R
7	The kVA rating of an isolation transformer increases when connected as an auto-transformer due to	M4.04	R
8	In a given transformer for given applied voltage ,losses which remain constant irrespective of load change are ----- losses	M4.04	R
9	It is advisable to use auto-transformer if the transformation ratio is	M4.04	R

PART B

IV. Answer any **eight** questions from the following, each question carries **3 marks**.
(8 * 3=24 marks)

1	What are the effects of armature reaction on the performance of dc machine	M1.01	U
---	--	-------	---

2	Dummy coils are not used in lap winding. Justify with suitable example	M1.03	U
3	Draw the mechanical characteristics of DC shunt motor	M2.04	U
4	What is a starter? What is the necessity of starter in dc motor	M2.01	R
5	Differentiate between core and shell type transformer(any 5)	M3.01	U
6	Derive the emf equation of single phase transformer	M3.02	A
7	Illustrate the phasor diagrams of transformers	M3.03	U
8	Explain the working principle of an autotransformer	M4.03	U
9	What are the advantages and disadvantages of autotransformer?	M4.03	R
10	What are the advantages of delta-delta connection	M4.04	R

PART C

Answer **ALL** questions. Each question carries **7 marks**.

(6*7=42 marks)

III	Explain the construction of a DC machine with neat diagram.	M1.01	R
	OR		
IV	Explain different types of DC generator with neat circuit diagram and necessary equations.	M1-02	R
V	Draw the circuit diagram and explain the experimental procedure to conduct Hopkinson test on DC machine	M2-01	A
	OR		
VI	A four pole generator having wave-wound armature winding has 51 slots, each slot containing 20 conductors. What will be the voltage generated in the machine when driven at 1500 rpm assuming the flux per pole to be 7.0 mWb ?	M2-02	A
VII	Explain methods of speed control for d.c.motors	M2-03	R
	OR		
VIII	Explain four point starter used for d.c.shunt motor	M2-03	R
IX	Draw the equivalent circuit of the 1-phase transformer	M3-04	R
	OR		
X	Explain the short circuit test in 1-phase transformer	M4-01	R

XI	Derive the condition for maximum efficiency and the load current at which max. Efficiency occurs in a single phase transformer	M4-01	A
	OR		
XII	Discuss the operation of open delta (V-V) configuration of transformers	M4-04	R
XIII	Explain regulation and its causes in transformer.	M4-01	R
	OR		
XIV	Explain the various connections for 3-phase transformer	M4-04	R

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IV	15	31.46	3		3		5		11	
			-----	3	-----	9	-----	35	-----	47
Total			9		10		12		31	
			-----	9	-----	30	-----	84	-----	123

Cognitive Level Wise Question Analysis

level	% mark distribu tion	mark s/leve l	Type of questions							
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R	30	36.9	9		3		9		21	
			-----	9	-----	9	-----	63	-----	72
U	50	61.5	0		6		0		6	
			-----	0	-----	18	-----	0	-----	18
A	20	24.6	0		1		3		4	
			-----	0	-----	3	-----	21	-----	24
Total			9		10		12		31	
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Question Wise Analysis

Q.No	Module Outcome	Cognitive Level	Marks	Time
I.1	M1.01	R	1	2
I.2	M1.03	R	1	2
I.3	M1.04	R	1	2
I.4	M2.01	R	1	2
I.5	M2.02	R	1	2
I.6	M3.01	R	1	2
I.7	M3.02	R	1	2
I.8	M3.03	R	1	2
I.9	M4.01	R	1	2
II.1	M1.02	U	3	7
II.2	M1.01	U	3	7
II.3	M1.01	U	3	8
II.4	M1.03	R	3	8
II.5	M2.03	U	3	7
II.6	M3.03	A	3	8
II.7	M3.04	U	3	7
II.8	M3.02	U	3	7
II.9	M3.02	R	3	7
II.10	M4.02	R	3	7
III.	M1.04	R	7	17
IV.	M1.03	R	7	17
V	M2.01	R	7	17
VI	M2.04	R	7	17
VII	M2.02	R	7	17

VIII	M2.02	R	7	17
IX	M3.02	R	7	17
X	M3.02	R	7	17
XI	M4.03	A	7	17
XII	M4.03	R	7	17
XIII	M4.04	R	7	17
XIV	M4.04	R	7	17
Total			123	295

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Scoring Indicators

Model Question Paper I

- 1.DC Generator
- 2.Commutator
- 3.A=2
- 4.Dc series motor
- 5.Dangerously high speed
- 6.n=5.5
- 7.establishment of conductive link between the primary and secondary.
- 8.cold-rolled grain oriented (CRGO) steel
- 9.Voltage transformation ratio

PART B

1. Let Φ = flux/pole in Wb, Z = total number of armature conductors, P = number of poles

A = number of parallel paths = 2 ... for wave winding

= P ... for lap winding

N = speed of armature in r.p.m.

E_g = e.m.f. of the generator = e.m.f./parallel path

Flux cut by one conductor in one revolution of the armature,

$d\phi = P\phi$ webers

Time taken to complete one revolution,

$dt = 60/N$ second

e.m.f generated/conductor = $d\phi/dt = P\phi / \{60/N\} = P\phi N/60$

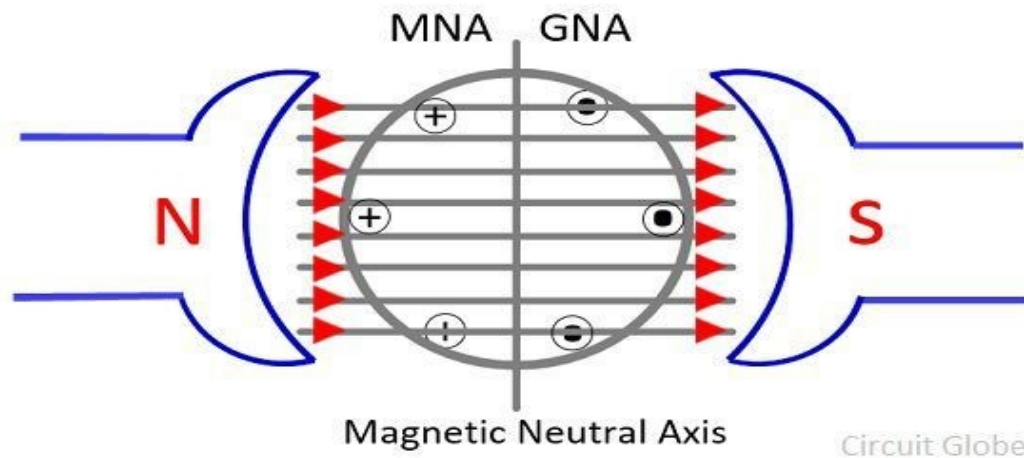
E_g = e.m.f. per parallel path = (e.m.f/conductor) ' No. of conductors in series per parallel path.

$$E_g = P\phi N/60 * (Z/A)$$

2.The significance of Back EMF

Back emf is very significant in the working of a dc motor. The presence of back emf makes the d.c. motor a self-regulating machine i.e., it makes the motor to draw as much armature current as is just sufficient to develop the torque required by the load.

3.The armature reaction simply shows the effect of armature field on the main field. In other words, the armature reaction represents the impact of the armature flux on the main field flux. The armature field is produced by the armature conductors when current flows through them. And the main field is produced by the magnetic poles.



4. A series motor should never be started at no load. With no mechanical load on the series motor, the current is low, the counter-EMF produced by the field winding is weak, and so the armature must turn faster to produce sufficient counter-EMF to balance the supply voltage. The motor can be damaged by overspeed

5. Commutation in DC machines is the process by which the reversal of current takes place. In DC generator this process is used to convert the induced AC in the conductors to a DC output. Each Armature coil contains two commutators attached at its end. For the transformation of current, the Commutator segments and brushes should maintain a continuously moving contact. To get larger output values more than one coil is used in DC machines. So, instead of one pair, we have a number of pairs of Commutator segments.

6. Load characteristics of DC shunt motor

- The equation for back emf induced in a dc motor is given as $E_b = P\phi NZ / 60A$
- In a dc shunt motor, the field is connected across the armature terminals
- For a dc motor, the expression for torque is given as $T = K \phi I_a$ i.e., torque is directly proportional to flux and armature current

7. The transformer will give the maximum efficiency when their copper loss is equal to the iron loss. A loss in a transformer that takes place in the winding resistance of a transformer is known as the copper loss

8.All-day efficiency of the transformer depends on their load cycle. The load cycle of the transformer means the repetitions of load on it for a specific period. The ordinary or commercial efficiency of a transformer is defined as the ratio of the output power to the input power

9.Three phase transformer connections In three phase system, the three phases can be connected in either star or delta configuration. In case you are not familiar with those configurations, study the following image which explains star and delta configuration. In any of these configurations, there will be a phase difference of 120° between any two phases

- **Star-Star (Y-Y)**
- **Delta-Delta (Δ - Δ)**
- **Star-Delta OR Wye-Delta (Y- Δ)**
- **Delta-Star OR Delta-Wye (Δ -Y)**

Star-Star (Y-Y)

- Star-star connection is generally used for small, high-voltage transformers. Because of star connection, number of required turns/phase is reduced (as phase voltage in star connection is $1/\sqrt{3}$ times of line voltage only). Thus, the amount of insulation required is also reduced.
- The ratio of line voltages on the primary side and the secondary side is equal to the transformation ratio of the transformers.
- Line voltages on both sides are in phase with each other.
- This connection can be used only if the connected load is balanced.

Delta-Delta (Δ - Δ)

- This connection is generally used for large, low-voltage transformers. Number of required phase/turns is relatively greater than that for star-star connection.
- The ratio of line voltages on the primary and the secondary side is equal to the transformation ratio of the transformers.
- This connection can be used even for unbalanced loading.

- Another advantage of this type of connection is that even if one transformer is disabled, system can continue to operate in open delta connection but with reduced available capacity.

Star-Delta OR Wye-Delta (Y-Δ)

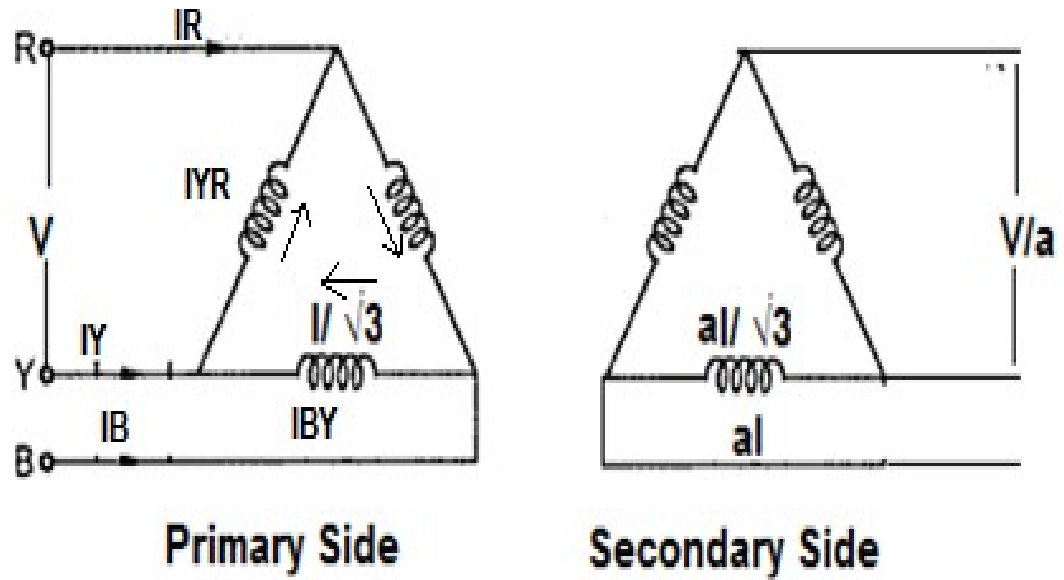
- The primary winding is star star (Y) connected with grounded neutral and the secondary winding is delta connected.
- This connection is mainly used in step down transformer at the substation end of the transmission line.
- The ratio of secondary to primary line voltage is $1/\sqrt{3}$ times the transformation ratio.
- There is 30° shift between the primary and secondary line voltages.

Delta-Star OR Delta-Wye (Δ-Y)

- The primary winding is connected in delta and the secondary winding is connected in star with neutral grounded. Thus it can be used to provide 3-phase 4-wire service.
- This type of connection is mainly used in step-up transformer at the beginning of transmission line.
- The ratio of secondary to primary line voltage is $\sqrt{3}$ times the transformation ratio.
- There is 30° shift between the primary and secondary line voltages

10. Disadvantage of Delta-Delta Connection:

- Due to the absence of neutral point it is not suitable for three phase four wire system.
- More insulation is required and the voltage appearing between windings and core will be equal to full line voltage in case of earth fault on one phase



PART C

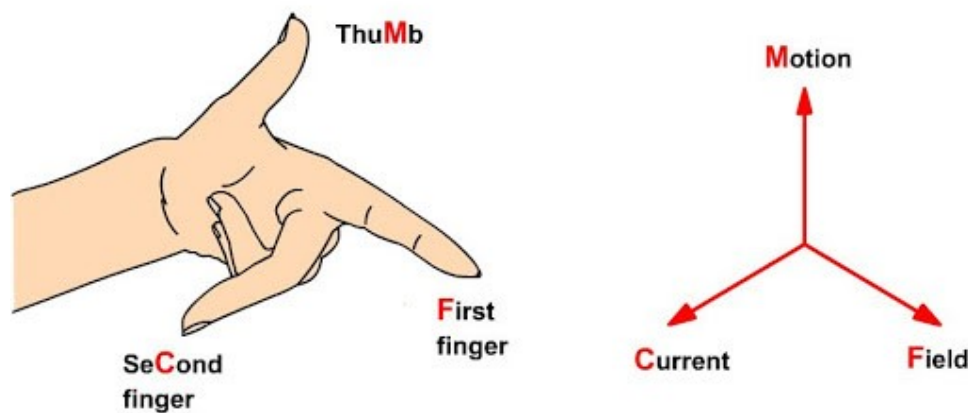
III.A DC motor is an electrical machine which converts electrical energy into mechanical energy. The basic working principle of the DC motor is that whenever a current carrying conductor places in the magnetic field, it experiences a mechanical force.



Fleming's left-hand rule and its magnitude decide the direction of this force.

Fleming's Left Hand Rule:

If we stretch the first finger, second finger and thumb of our left hand to be perpendicular to each other, and first finger represents the direction of the magnetic field, the second finger represents the direction of the current, then the thumb represents the direction of the force experienced by the current carrying conductor.



$$F = BIL \text{ Newtons}$$

Where,

B = magnetic flux density,

I = current and

L = length of the conductor within the magnetic field.

When armature winding is connected to a DC supply, an electric current sets up in the winding. Permanent magnets or field winding (electromagnetism) provides the magnetic field. In this case, current carrying armature conductors experience a force due to the magnetic field, according to the principle stated above.

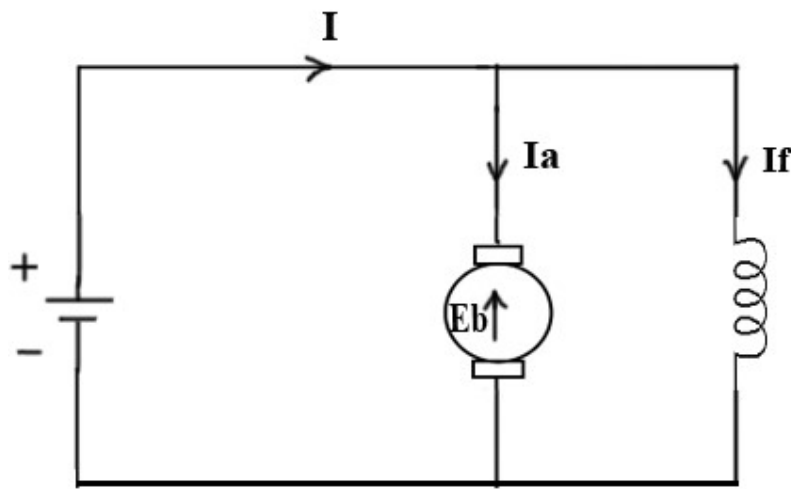
The Commutator is made segmented to achieve unidirectional torque. Otherwise, the direction of force would have reversed every time when the direction of movement of the conductor is reversed in the magnetic field. This is how a DC motor works!

Back- EMF of DC motor

According to the fundamental law of nature, no energy conversion is possible until there is something to oppose the conversion. In case of generators, magnetic drag provides this opposition, but in the case of dc motors, there is back emf. Presence of the back emf makes a dc motor 'self-regulating'.

When the armature of a motor is rotating, the conductors are also cutting the magnetic flux lines and hence according to the Faraday's law of electromagnetic induction, an emf induces in the armature conductors.

The direction of this induced emf is such that it opposes the armature current (I_a). The circuit diagram below illustrates the direction of the back emf and armature current.



Significance of Back-

EMF

Magnitude of back emf is directly proportional to speed of the motor. Consider the load on a dc motor is suddenly reduced. In this case, required torque will be small as compared to the current torque. Speed of the motor will start increasing due to the excess torque. Hence, being proportional to the speed, magnitude of the back emf will also increase. With increasing back emf armature current will start decreasing. Torque being proportional to the armature current, it will also decrease until it becomes sufficient for the load. Thus, speed of the motor will regulate.

On the other hand, if a dc motor is suddenly loaded, the load will cause decrease in the speed. Due to decrease in speed, back emf will also decrease which allows more armature current. Due to increase in armature current the torque will increase to fulfill the load requirement.

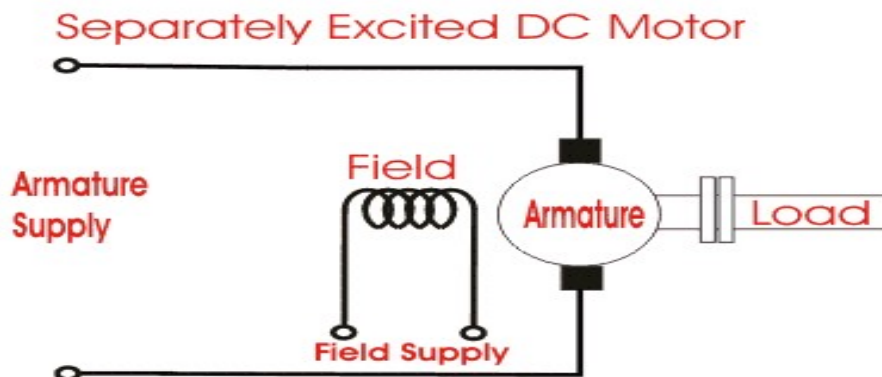
IV. There are three methods of excitation, and thus three main types of DC generators:

1. Permanent Magnet DC Generators – Field coils excited by permanent magnets
2. Separately Excited DC Generators – Field coils excited by some external source
3. Self Excited DC Generators – Field coils excited by the generator itself. Self-excited DC generators can further be classified depending on the position of their field coils. The three types of self-excited DC generators are:
 1. Series Wound Generators
 2. Shunt Wound Generators
 3. Compound Wound Generator

V. The working of the three-point starter can be stated as follows –

- Initially, the DC supply is switched on with handle in the OFF position.
- The handle is now moved to the first stud. When it comes in contact with the first stud, the whole starting resistance is inserted in series with the armature winding and the shunt field winding is directly connected across the DC supply.
- As the handle is gradually moved over to the final stud, the starting resistance is cut out from the armature circuit in steps. After reaching to the final stud, the handle is held magnetically by the NVC which is energized by the shunt field current.
- If the supply voltage is interrupted or if an open-circuit is occurred in the field circuit, the NVC is de-energized and the handle goes back to the OFF position under the pull of the control spring. If the NVC were not used, then in case of failure of supply, the handle would remain in contact with the final stud. When the supply is restored, the motor will be directly connected across the full supply voltage, resulting in an excessive armature current and may damage the motor.
- If the motor is overloaded or if a short circuit is occurred, it will draw a large current from the supply. This excessive current will increase the mmf of the OLC and pull the plunger P, which short-circuits the NVC. Hence, the NVC is de-energised and the handle is pulled to the OFF position by the control spring. Therefore, the motor is automatically isolated from the supply.

VI. In separately excited DC motor, the main supply is given separately to the armature and field windings. The major distinguishing fact in these forms of DC motor is that the armature current does not move across the field windings as the field winding is powered from a separate external supply of DC. From the torque equation of motors, we know that:



So, the torque, in this case, can be changed by modifying the field flux (ϕ), independent of the current of the armature (I_a). The figure below presents the separately excited DC motor.

The schematic cross-section across a two-pole DC motor is presented in the next figure, including the fixed stator S and the cylindrical rotor, introduced as armature A. Once rotor and pole shoes are always connected in order to decrease the iron wastes caused by the varying magnetic flux, the rest of the stator is laminated just in huge machines, when the device is needed to perform with fast varying torque and speed or once a static power converter with greatly distorted currents and voltages is used as the power supply.

The basic poles (M and P) are connected with the field windings, moving the field current i_f which runs the main flux across the rotor and stator. A closed armature coil is set in the axial slots of the rotor and combined with the commutator bars; it is provided by the brushes and the commutator produces the armature current i_a . This generates a distributed ampere-turn

(mmf) wave, fixed in the space and rotated in the direction of the quadrature axis, orthogonal to the basic axis, so that maximum output torque for the armature current can be created.

In the case of the great distance in the quadrature direction, the resulting armature flux is much smaller than the basic flux. It can be decreased even further by setting compensating coils in the axial slots on the pole shoes and arranging them in series with the armature. Their opposing ampere-turn stops the quadrature field induced by the armature and eliminates the undesired armature reaction, which otherwise wants to distort the even distribution of the main flux under the poles across the circumference of the rotor.

VII The weight of the copper is proportional to the length and area of a cross-section of the conductor. The length of the conductor is proportional to the number of turns, and the cross-section is proportional to the product of current and number of turns. Now, from the above figure (B) shown of the auto transformer, the weight of copper required in an auto transformer is

$W_a = \text{weight of copper in section AC} + \text{weight of copper in section CB}$

Therefore

$$W_a \propto I_1 (N_1 - N_2) + (I_2 - I_1) N_2$$

$$W_a \propto I_1 N_1 + I_2 N_2 - 2I_1 N_2$$

If the same duty is performed with an ordinary two winding transformer shown above in the figure (A), the total weight of the copper required in the ordinary transformer,

$W_0 = \text{weight of copper on its primary winding} + \text{weight of copper on its secondary winding}$

Therefore,

$$W_0 \propto I_1 N_1 + I_2 N_2$$

Now, the ratio of the weight of the copper in an auto transformer to the weight of copper in an ordinary transformer is given as

$$\frac{W_a}{W_o} = \frac{I_1 N_1 + I_2 N_2 - 2I_1 N_2}{I_1 N_1 + I_2 N_2}$$

OR

$$\frac{W_a}{W_o} = \frac{I_1 N_1 + I_2 N_2}{I_1 N_1 + I_2 N_2} - \frac{2I_1 N_2}{I_1 N_1 + I_2 N_2}$$

$$\frac{W_a}{W_o} = 1 - \frac{2I_1 N_2 / I_1 N_1}{I_1 N_1 / I_1 N_1 + I_2 N_2 / I_1 N_1} = 1 - K$$

OR

$$W_a = (1 - K)W_o$$

Saving of copper affected by using an auto transformer = weight of copper required in an ordinary transformer – weight of copper required in an auto transformer

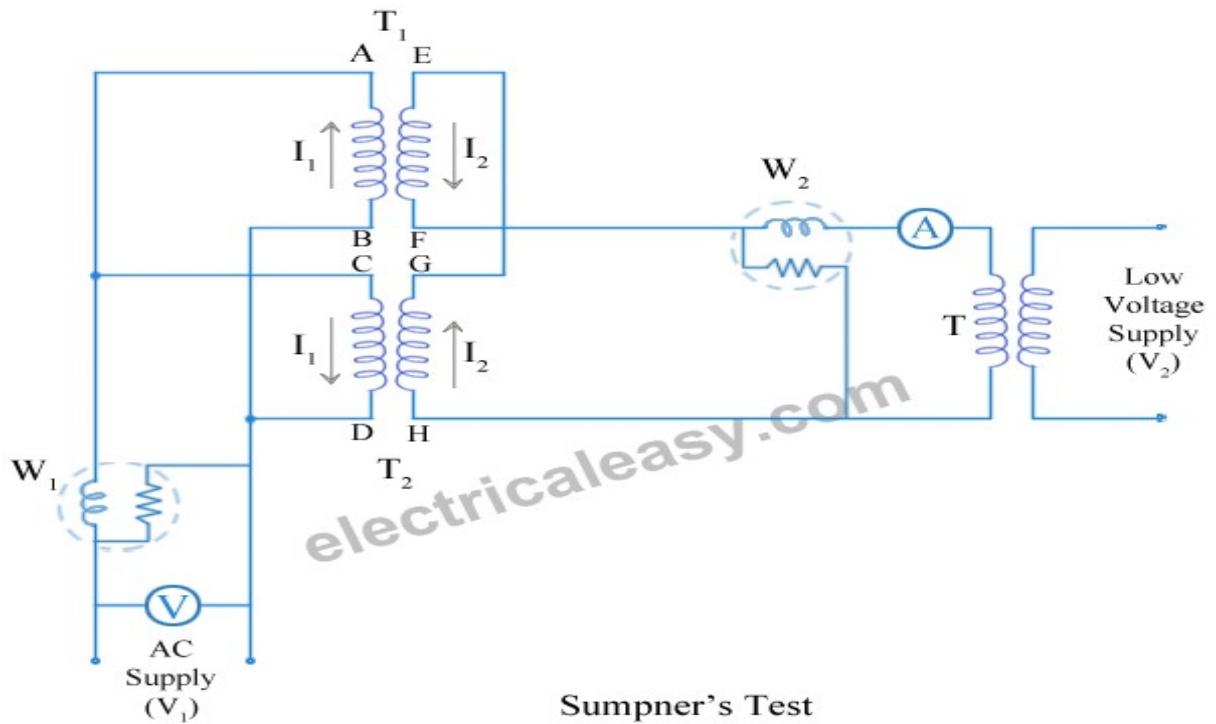
$$\text{Saving of copper} = W_o - W_a = W_o - (1 - K)W_o = KW_o$$

Therefore,

Saving of copper = K x weight of copper required for two windings of the transformer

Hence, saving in copper increases as the transformation ratio approaches unity. Hence the auto transformer is used when the value of K is nearly equal to unity.

VIII. Sumpner's test or back to back test can be employed only when two identical transformers are available. Both transformers are connected to supply such that one transformer is loaded on another. Primaries of the two identical transformers are connected in parallel across a supply. Secondaries are connected in series such that emf's of them are opposite to each other. Another low voltage supply is connected in series with secondaries to get the readings, as shown in the circuit diagram shown below.



In above diagram, T_1 and T_2 are identical transformers. Secondaries of them are connected in voltage opposition, i.e. E_{EF} and E_{GH} . Both the emf's cancel each other, as transformers are identical. In this case, as per superposition theorem, no current flows through secondary. And thus the no load test is simulated. The current drawn from V_1 is $2I_0$, where I_0 is equal to no load current of each transformer. Thus input power measured by wattmeter W_1 is equal to iron losses of both transformers.

i.e. iron loss per transformer $P_i = W_1/2$.

Now, a small voltage V_2 is injected into secondary with the help of a low voltage transformer. The voltage V_2 is adjusted so that, the rated current I_2 flows through the secondary. In this case, both primaries and secondaries carry rated current. Thus short circuit test is simulated and wattmeter W_2 shows total full load copper losses of both transformers. i.e. copper loss per

transformer $P_{Cu} = W_2/2$. From above test results, the **full load efficiency of each transformer** can be given as -

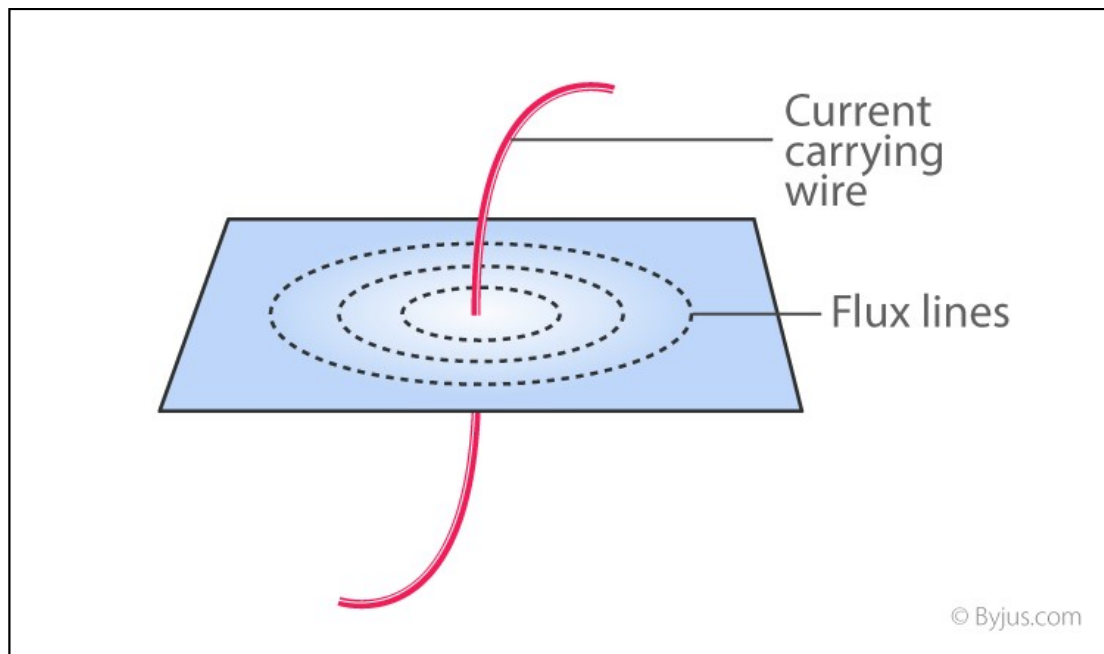
$$\text{\% full load efficiency of each transformer} = \frac{\text{output}}{\text{output} + \frac{W_1}{2} + \frac{W_2}{2}} \times 100$$

IX. A Scott-T transformer (also called a Scott connection) is a type of circuit used to derive two-phase power from a three-phase source or vice-versa. The Scott connection **evenly distributes a balanced load between the phases of the source**. ... The Main and Teaser outputs are 90 degrees out of phase.

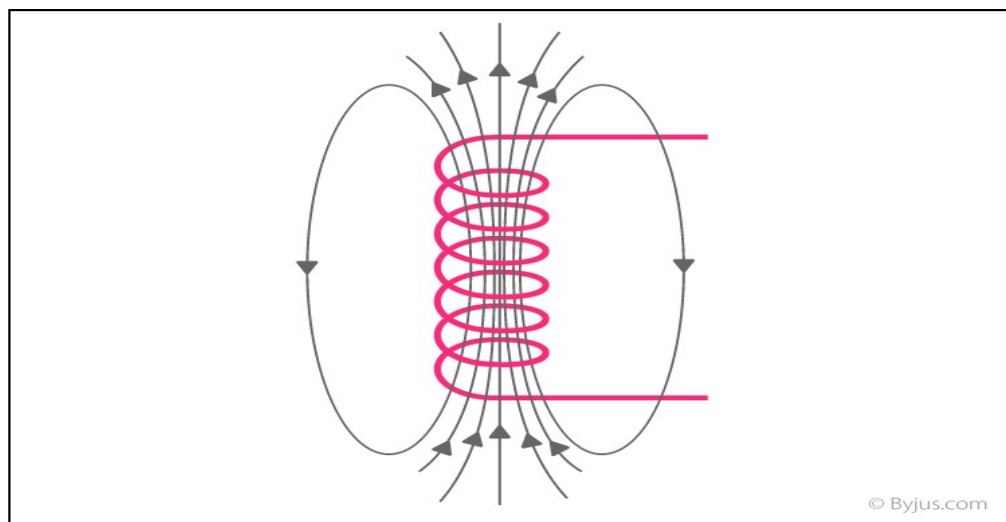
X. The transformer works on the principle of Faraday's law of electromagnetic induction and mutual induction. There are usually two coils primary coil and secondary coil on the transformer core. The core laminations are joined in the form of strips. The two coils have high mutual inductance. When an alternating current pass through the primary coil it creates a varying magnetic flux. As per faraday's law of electromagnetic induction, this change in magnetic flux induces an emf (electromotive force) in the secondary coil which is linked to the core having a primary coil. This is mutual induction.

Overall, a transformer carries the below operations:

1. Transfer of electrical energy from circuit to another
2. Transfer of electrical power through electromagnetic induction
3. Electric power transfer without any change in frequency
4. Two circuits are linked with mutual induction

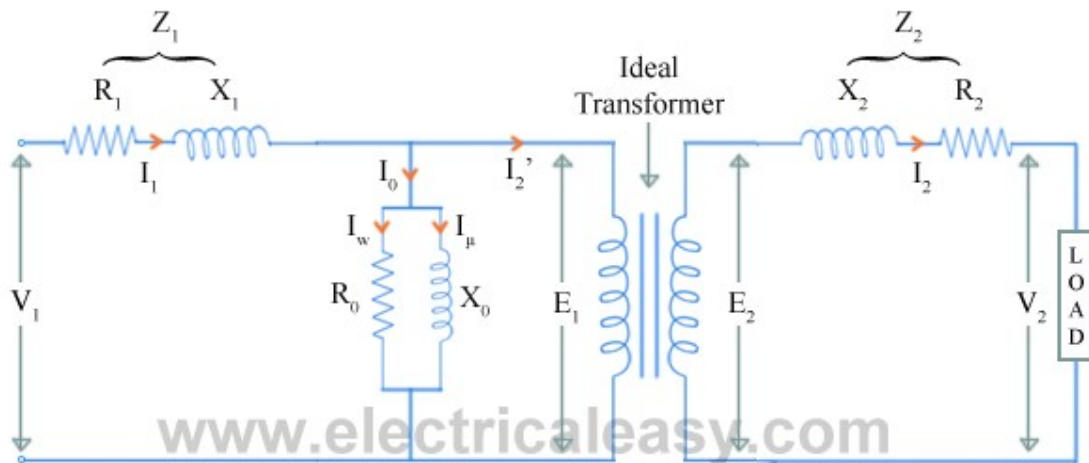


The figure shows the formation of magnetic flux lines around a current-carrying wire. The normal of the plane containing the flux lines are parallel to normal of a cross-section of a wire.



The figure shows the formation of varying magnetic flux lines around a wire-wound. The interesting part is that reverse is also true, when a magnetic flux line fluctuates around a piece of wire, a current will be induced in it.

XI. Resistances and reactances of transformer, which are described above, can be imagined separately from the windings (as shown in the figure below). Hence, the function of windings, thereafter, will only be the transforming the voltage.



The no load current I_0 is divided into, pure inductance X_0 (taking magnetizing components I_μ) and non induction resistance R_0 (taking working component I_w) which are connected into parallel across the primary. The value of E_1 can be obtained by subtracting $I_1 Z_1$ from V_1 . The value of R_0 and X_0 can be calculated as, $R_0 = E_1 / I_w$ and $X_0 = E_1 / I_\mu$. But, using this equivalent circuit does not simplify the calculations. To make calculations simpler, it is preferable to transfer current, voltage and impedance either to primary side or to the secondary side. In that case, we would have to work with only one winding which is more convenient.

From the voltage transformation ratio, it is clear that, $E_1 / E_2 = N_1 / N_2 = K$

Now, let's refer the parameters of secondary side to primary.

Z_2 can be referred to primary as Z_2'

where, $Z_2' = (N_1/N_2)^2 Z_2 = K^2 Z_2$where $K = N_1/N_2$.

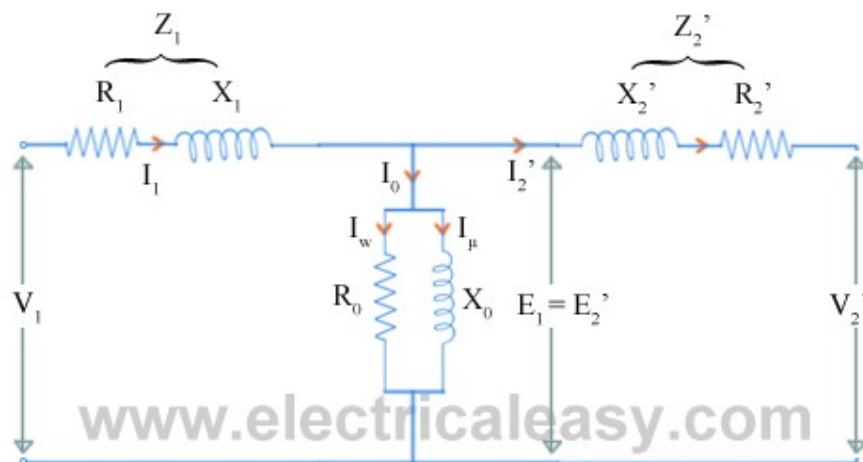
that is, $R_2' + jX_2' = K^2(R_2 + jX_2)$

equating real and imaginary parts,

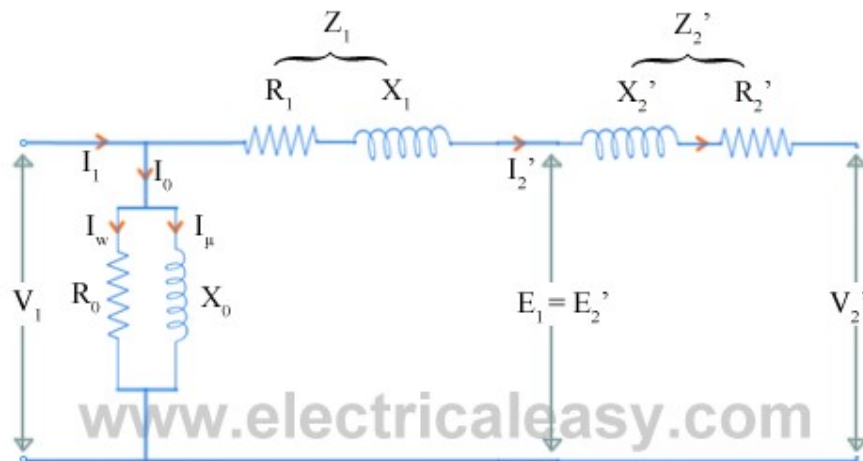
$R_2' = K^2 R_2$ and $X_2' = K^2 X_2$.

And $V_2' = KV_2$

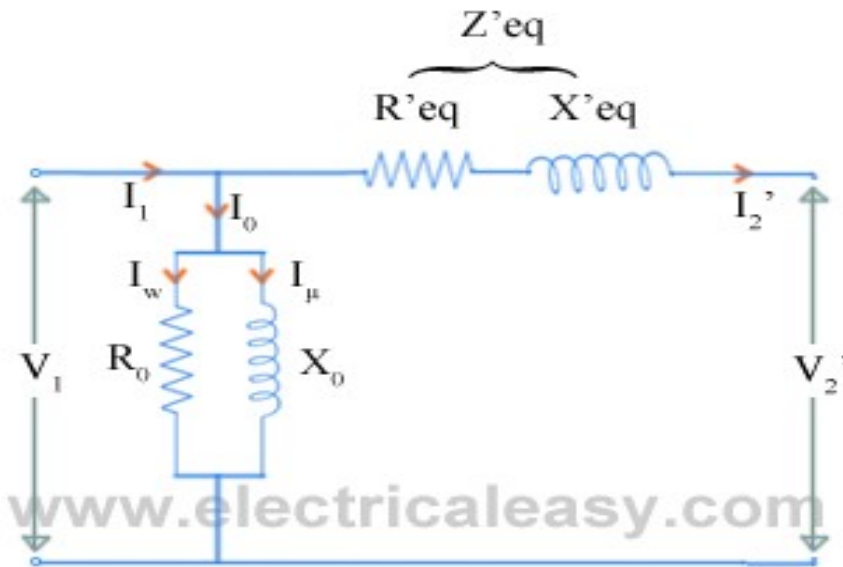
The following figure shows the **equivalent circuit of transformer with secondary parameters referred to the primary.**



Now, as the values of winding resistance and leakage reactance are so small that, V_1 and E_1 can be assumed to be equal. Therefore, the exciting current drawn by the parallel combination of R_0 and X_0 would not affect significantly, if we move it to the input terminals as shown in the figure below.

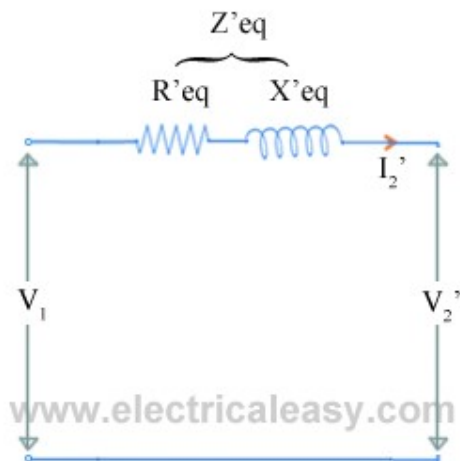


Now, let $R_1 + R_2' = R_{eq}$ and $X_1 + X_2' = X_{eq}$. Then the **equivalent circuit of transformer** becomes as shown in the figure below



Approximate Equivalent Circuit Of Transformer

If only voltage regulation is to be calculated, then even the whole excitation branch (parallel combination of R_0 and X_0) can be neglected. Then the equivalent circuit becomes as shown in the figure below



XII. Working Principle of Ideal Transformer

An ideal transformer works on two principles like when an electric current generates a magnetic field and a changing magnetic field in a coil induces a voltage across the coil ends. When the current is changed within the primary coil, then the magnetic flux is developed. So changing magnetic field can induce a voltage within the secondary coil.

When the current flows through the primary coil then it creates a magnetic field. The two windings are wrapped in the region of a very high magnetic core like iron, so the magnetic flux supplies through the two windings. Once a load is connected to the secondary coil, then the voltage and current will be in the indicated direction.

The **properties of an ideal transformer** include the following.

- The two windings of this transformer have small resistance.
- Because of the resistance, eddy current and hysteresis there are no losses in the transformer.
- The efficiency of this transformer is 100%
- The total flux generated in the transformer has restricted the core & connects with the windings. Therefore, its flux & inductance leakage is zero.
- The core has unlimited permeability so a negligible magnetomotive force is necessary to arrange the flux within the core.

An ideal transformer model is shown below. This transformer is ideal in three conditions when it has no leakage flux, no windings resistance and no iron loss within the core. The properties of practical as well as ideal transformers are not similar to each other.

XIII. A transformer that is used to measure electrical quantities like current, voltage, power, frequency and power factor is known as an instrument transformer. These transformers are mainly used with relays to protect the power system



instrument-transformer

The **Purpose of the instrument transformer** is to step down the voltage & current of the AC system because the level of voltage & current in a power system is extremely high. So designing the measuring instruments with high voltage & current is difficult as well as expensive. In general, these instruments are mainly designed for 5 A & 110 V.

The measurement of high-level electrical quantities can be done using a device namely instrument transformer. These transformers play an essential role in current power systems.

Types of Instrument Transformers

Instrument transformers are classified into two types such as

- Current Transformer
- Potential Transformer

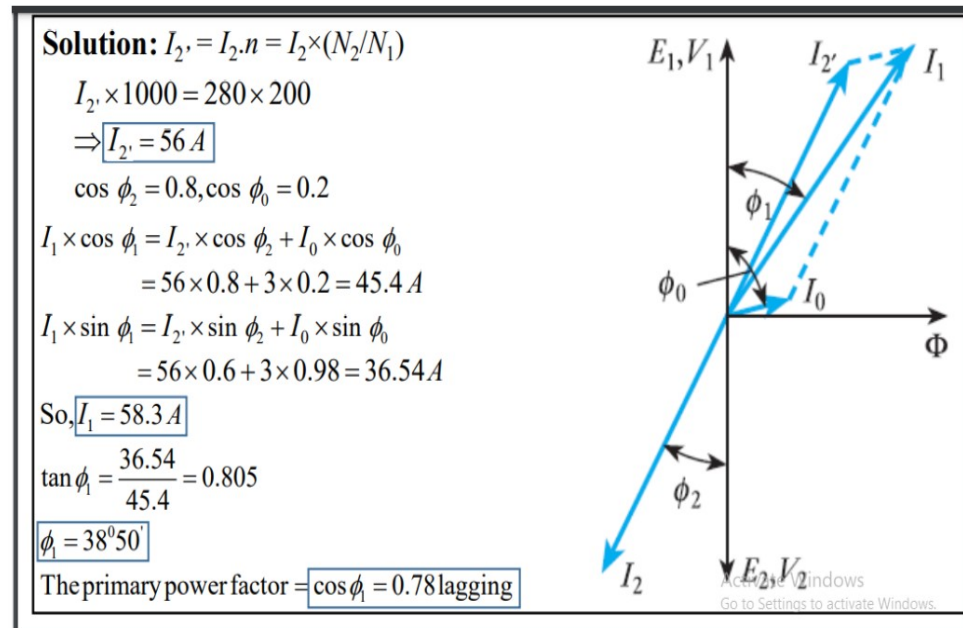
Current Transformer

This type of transformer can be used in power systems to step down the voltage from a high level to a low level with the help of a 5A ammeter. This transformer includes two windings like primary and secondary. The current in the secondary winding is proportional to the current in the primary winding as it generates current in the secondary winding. The circuit diagram of a typical current transformer is demonstrated in the following figure.

Current-transformer: In this transformer, the primary winding consists of few turns and it is connected with the power circuit in series. So it is called a series transformer. Likewise, the secondary winding includes a number of turns and it is connected to an ammeter directly because the ammeter includes small resistance.

Thus, the secondary winding of this transformer works almost in the condition of a short circuit. This winding includes two terminals where one of its terminals is connected to ground to evade the huge current. So insulation breakdown chances will be reduced to guard the operator from huge voltage. The secondary winding of this transformer in the above circuit is short-circuited before disconnecting the ammeter with the help of a switch to avoid the high voltage across the winding.

XIV.



Scoring Indicators

Model Question Paper II

PART A

1. Carbon / Graphite
2. load Current
3. The torque required
4. in this case field coils carry the load or armature current. As the cross-sectional area is large, the resistance is minimum and series voltage drop is also minimum
5. no core loss, no ohmic resistance, and no leakage flux.
6. kVA
7. a
8. Hysteresis and eddy current losses
9. near to 1

PART B

1. The adverse effects of armature reaction:
 - Armature reaction weakens the main flux. In case of a dc generator, weakening of the main flux reduces the generated voltage.
 - Armature reaction distorts the main flux, hence the position of M.N.A. gets shifted (M.N.A. is perpendicular to the flux lines of main field flux). Brushes should be placed on the M.N.A., otherwise, it will lead to sparking at the surface of brushes. So, due to armature reaction, it is hard to determine the exact position of the MNA
2. The dummy coils are similar to other coils except their ends are cut, short and taped. They do not connect with the commutator bars. The dummy coils are **simply to provide mechanical balance for the armature**. As they do not connect with commutator bars, they do not affect the electrical characteristics of the winding.
3. Generally, three characteristic curves are considered important for DC motors which are,
(i) Torque vs. armature current,

(ii) Speed vs. armature current

(iii) Speed vs. torque.

These characteristics are determined by keeping the following two relations

$$T_a \propto \Phi \cdot I_a \text{ and } N \propto E_b / \Phi$$

These above equations can be studied at - emf and torque equation of dc machine.

For a DC motor, magnitude of the back emf is given by the same emf equation of a dc generator

i.e. $E_b = P\Phi NZ / 60A$. For a machine, P, Z and A are constant
therefore, $N \propto E_b / \Phi$

4. Starters are **used to protect DC motors from damage that can be caused by very high current and torque during startup**. They do this by providing external resistance to the motor, which is connected in series to the motor's armature winding and restricts the current to an acceptable level

Necessity of starter

When the motor is in off condition, the armature is stationary and the back EMF which is proportional to speed is also zero. In Motor armature resistance is very low. If the rated voltage is applied to the armature, it will draw heavy current many times that of full load current. And there is more possibility of damaging the armature due to heavy starting current.

So, we need to limit the starting current to its safe value. This is possible by inserting a resistance in series with the armature at the time of starting for a short period. As the motor gains rated speed, back EMF is built up in it, and then the starting resistance could be gradually cut off in step by step.

5. Any 5 comparison

Basis Comparison	Core Type Transformer	Shell Type Transformer
Definition	The winding surround the core.	The core surround the winding.
Lamination Shape	The lamination is cut in the form of the L strips.	Lamination are cut in the form of the long strips of E and L.

Cross Section	Cross-section may be square, cruciform and three stepped	The cross section is rectangular in shape.
Copper Require	More	Less
Other Name	Concentric Winding or Cylindrical Winding.	Sandwich or Disc Winding
Limb	Two	Three
Insulation	More	Less
Flux	The flux is equally distributed on the side limbs of the core.	Central limb carry the whole flux and side limbs carries the half of the flux.
Winding	The primary and secondary winding are placed on the side limbs.	Primary and secondary windings are placed on the central limb
Magnetic Circuit	Two	One
Losses	More	Less
Maintenance	Easy	Difficult
Mechanical Strength	Low	High

Output	Less	High
Natural Cooling	Does not Exist	Exist

6. When an alternating voltage V_1 is applied to the transformer primary side, an alternating flux will set up in the core which links both primary and secondary winding. This will result in self induced emf (E_1) sets up in the transformer primary winding and mutually induced emf (E_2) sets up in the secondary winding.

Let N_1 = Primary winding turns

N_2 = Secondary winding turns,

Φ_m = Maximum flux in the core (Weber)

$$= B_m \times A_i$$

B_m = Maximum flux density in the core (Weber / meter²)

A_i = Core area (Meter²)

F = Frequency of alternating supply (Hz)

- An alternating flux increases from zero to positive maximum in the one fourth of a cycle i.e. $T/4$ second
- According to Faraday's law of electro – magnetic induction, the magnitude of induced emf is directly proportional to rate of change of flux linkages.

$$E \propto d (N\Phi_m) / dt$$

$$\propto N (d\Phi_m) / dt, \text{ Where } N\Phi_m = \text{Flux linkages}$$

$$d\Phi_m = \text{Change in flux, } dt = \text{Change in time}$$

- Therefore the average value of induced emf per turn is given by

$$E = (\Phi_m - 0) / (T / 4)$$

$$= 4f \Phi_m$$

- As the flux varies sinusoidally, the induced emf will be sinusoidal.
- Form factor = RMS Value / Average Value

$$= 1.11 \dots\dots\dots(1)$$

(The value of form factor is 1.11 only for sinusoidal waveform)

RMS value of induced emf per turn = Form factor $\times$ Average emf

$$= 1.11 \times 4f \Phi_m$$

$$= 4.44f \Phi_m \dots\dots\dots(2)$$

RMS value of induced emf in the transformer primary winding

$$E_1 = 4.44f \Phi_m \times N_1$$

$$= 4.44f N_1 \Phi_m \dots\dots\dots(3)$$

Similarly the RMS value of induced emf in the transformer secondary winding

$$E_2 = 4.44f \Phi_m \times N_2$$

$$= 4.44f N_2 \Phi_m \dots\dots\dots(4)$$

As we know that $\Phi_m = B_m \times A_i$

$$E_1 = 4.44f B_m A_i N_1 \dots\dots\dots(5)$$

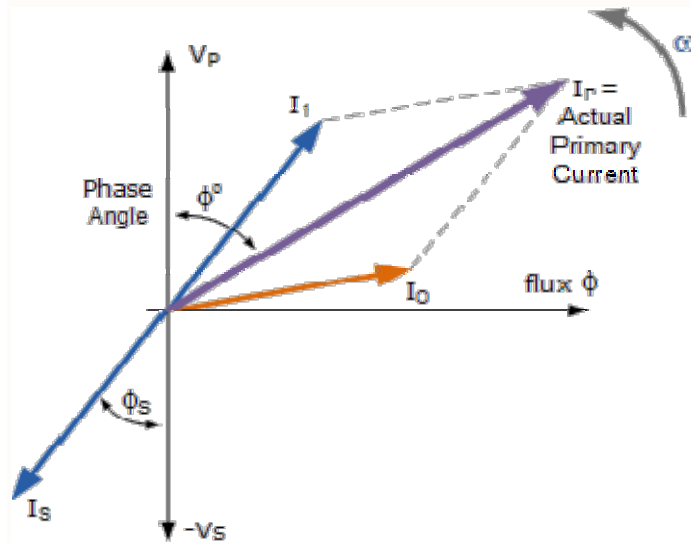
$$E_2 = 4.44f B_m A_i N_2 \dots\dots\dots(6)$$

From equation (5) and (6)

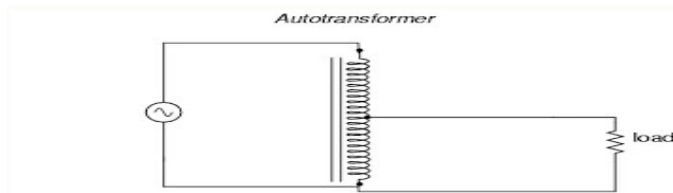
$$E_1 / E_2 = N_1 / N_2 \dots \dots \dots (7)$$

$$E_1 / N_1 = E_2 / N_2 = 4.44f B_m A_i \dots \dots \dots (8)$$

7.The total current drawn from the supply by the primary winding is the vector sum of the no-load current, I_0 and the additional supply current, I_1 as a result of the secondary transformer loading and which lags behind the supply voltage by an angle of Φ . We can show this relationship as a phasor diagram



8.consider the autotransformer circuit diagram shown below. As compared to two winding transformers as shown in Figure 1, Autotransformer has single winding. When an alternating supply is given to the primary circuit, because of Faraday's Law of electromagnetic Induction, an emf is induced in the primary part. Since the magnetic field is alternating in nature, and conductors are stationary.



The induced emf in primary produces a flux, which is called as primary winding flux. This flux links the secondary winding and induces an emf on secondary winding due to mutual induction. Hence emf is transferred in the secondary winding. Based on a number of turns on the secondary side, the magnitude of induced emf is determined

9. Advantages of autotransformer are given as follows:

- 1) An autotransformer required less Copper than an ordinary transformer of a similar rating.
- (2) It has better voltage regulation and operates at a higher efficiency than a 2-winding transformer of the same rating.
- (3) It has a smaller size than an ordinary transformer of the same rating.
- (4) An autotransformer requires a smaller exciting current than a 2-winding transformer of the same rating.

It may be noted that these advantages of the autotransformer decrease as the ratio of transformation increases. Therefore, an autotransformer has marked advantages only for relatively low values of transformation ratio

Disadvantages of autotransformer are given as follows:

- (1) There is a direct connection between the primary winding and the secondary. Therefore, the output is no longer directly isolated from the input.
- (2) An autotransformer is not safe for stepping down a high voltage to a low voltage. in case an open circuit develops in a common part of winding then full primary voltage appears across the load, it will dangerous to both people and equipment.

(3) The short-circuit current is much larger than for the two-winding transformer of the same rating

10. Advantages of delta connections:

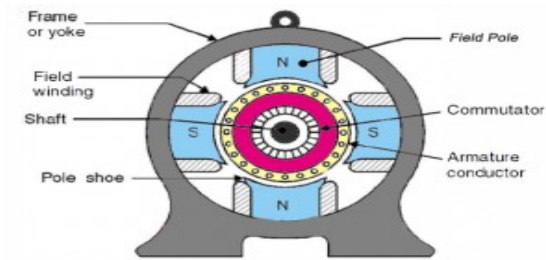
- More torque
- Efficient
- Simple motor design
- Heavy-duty application
- Protection is simple and less costly
- Used in rotatory conveyors
- While use delta connection, less current per winding for the same power output
- Major applications in power generation, transmission, and distribution
- The transformer secondary provides all the 3 phases
- Construction cost is low

PART C

III. A DC machine is an electromechanical energy alteration device. The working principle of a DC machine is when electric current flows through a coil within a magnetic field, and then the magnetic force generates a torque that rotates the dc motor. The DC machines are classified into two types such as DC generator as well as DC motor.

The main function of the DC generator is to convert mechanical power to DC electrical power, whereas a DC motor converts DC power to mechanical power. The AC motor is frequently used in industrial applications for altering electrical energy to mechanical energy. However, a DC motor is applicable where good speed regulation

Construction of DC Machine The construction of the DC machine can be done using some of the essential parts like Yoke, Pole core & pole shoes, Pole coil & field coil, Armature core, Armature winding otherwise conductor, commutator, brushes & bearings. Some of the parts of the DC machine



Yoke: Another name of a yoke is the frame. The main function of the yoke in the machine is to offer mechanical support intended for poles and protect the entire machine from moisture, dust, etc. The materials used in the yoke are designed with cast iron, cast steel otherwise rolled steel.

Pole and Pole Core: The pole of the DC machine is an electromagnet and the field winding is winding among pole. Whenever field winding is energized then the pole gives magnetic flux. The materials used for this are cast steel, cast iron otherwise pole core. It can be built with the annealed steel laminations for reducing the power drop because of the eddy currents.

Pole Shoe: Pole shoe in the DC machine is an extensive part as well as to enlarge the region of the pole. Because of this region, flux can be spread out within the air-gap as well as extra flux can be passed through the air space toward armature. The materials used to build pole shoe is cast iron otherwise cast steel, and also used annealed steel lamination to reduce the loss of power because of eddy currents.

Field Windings: In this, the windings are wound in the region of pole core & named as field coil. Whenever current is supplied through field winding than it electromagnetics the poles which generate required flux. The material used for field windings is copper.

Armature Core: Armature core includes a huge number of slots within its edge. The armature conductor is located in these slots. It provides the low-reluctance path toward the flux generated with field winding. The materials used in this core are permeability low-reluctance materials like iron otherwise cast. The lamination is used to decrease the loss because of the eddy current.

Armature Winding: The armature winding can be formed by interconnecting the armature conductor. Whenever an armature winding is turned with the help of prime mover then the

voltage, as well as magnetic flux, gets induced within it. This winding is allied to an exterior circuit. The materials used for this winding are conducting material like copper.

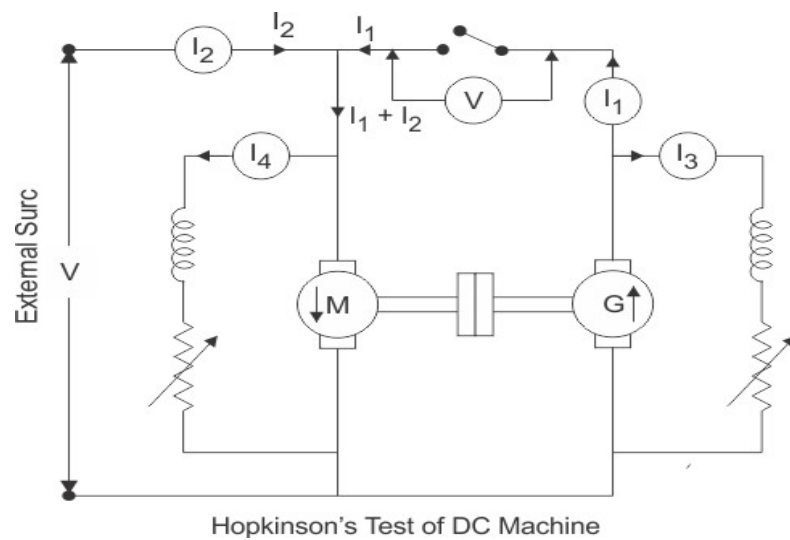
Commutator:The main function of the commutator in the DC machine is to collect the current from the armature conductor as well as supplies the current to the load using brushes. And also provides uni-directional torque for DC-motor. The commutator can be built with a huge number of segments in the edge form of hard drawn copper. The Segments in the commutator are protected from the thin mica layer.

Brushes:Brushes in the DC machine gather the current from the commutator and supply it to the exterior load. Brushes wear with time to inspect frequently. The materials used in brushes are graphite otherwise carbon which is in rectangular form.

IV. There are three methods of excitation, and thus three main types of DC generators:

4. Permanent Magnet DC Generators – Field coils excited by permanent magnets
5. Separately Excited DC Generators – Field coils excited by some external source
6. Self Excited DC Generators – Field coils excited by the generator itself. Self-excited DC generators can further be classified depending on the position of their field coils. The three types of self-excited DC generators are:
 4. Series Wound Generators
 5. Shunt Wound Generators
 6. Compound Wound Generator

V. Hopkinson's Test is another useful method of testing the efficiency of a DC machine. It is a full load test and it requires two identical machines which are coupled to each other. One of these two machines is operated as a generator to supply the mechanical power to the motor and the other is operated as a motor to drive the generator. For this process of back to back driving the motor and the generator, Hopkinson's test is also called back-to-back test or regenerative test. If there are no losses in the machine, then no external power supply would have needed. But due to the drop in the generator output voltage we need an extra voltage source to supply the proper input voltage to the motor. Hence, the power drawn from the external supply is therefore used to overcome the internal losses of the motor-generator set. Hopkinson's test is also called regenerative test or back to back test or heat run test.



Here is a circuit connection for the **Hopkinson's test** shown in figure below. A motor and a generator, both identical, are coupled together. When the machine is started it is started as motor. The shunt field resistance of the machine is adjusted so that the motor can run at its rated speed. The generator voltage is now made equal to the supply voltage by adjusting the shunt field resistance connected across the generator. This equality of these two voltages of generator and supply is indicated by the voltmeter as it gives a zero reading at this point connected across the switch. The machine can run at rated speed and at desired load by varying the field currents of the motor and the generator

V1. For a simplex wave wound generator,

$$Emf\ Generated / Path = \frac{\phi Z N}{60} \left(\frac{P}{A} \right)$$

$$E_g = \frac{P \phi Z N}{60 A}$$

Here,

$$\phi = 7 \times 10^{-3} \text{ Wb}$$

$$Z = 51 \times 20 = 1020$$

$$A = P = 4$$

$$N = 1500 \text{ rpm}$$

$\therefore$

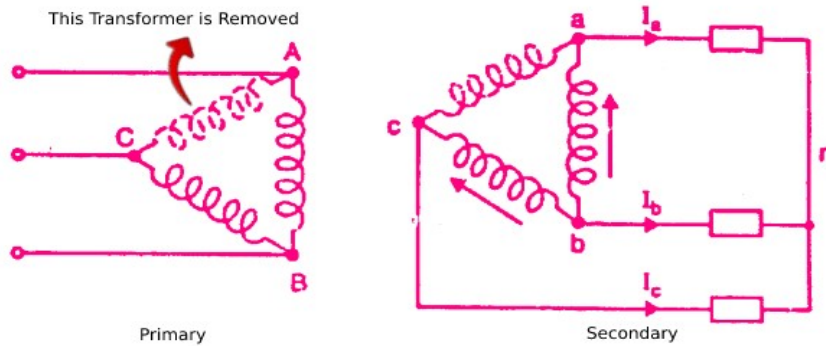
$$E_g = \frac{7 \times 10^{-3} \times 1020 \times 1500}{60} \left(\frac{4}{2} \right)$$

$$= 178.5 \text{ V}$$

VII. Open Delta Connection is a method of transformer connection used to transform three-phase power using two single-phase transformers. The Scott connection is another method used for the same purpose of transforming 3 phase power using two single-phase transformers. Both of these methods result in slightly unbalanced output voltages under load because of unsymmetrical relations. The unbalance is negligible under usual conditions of operation

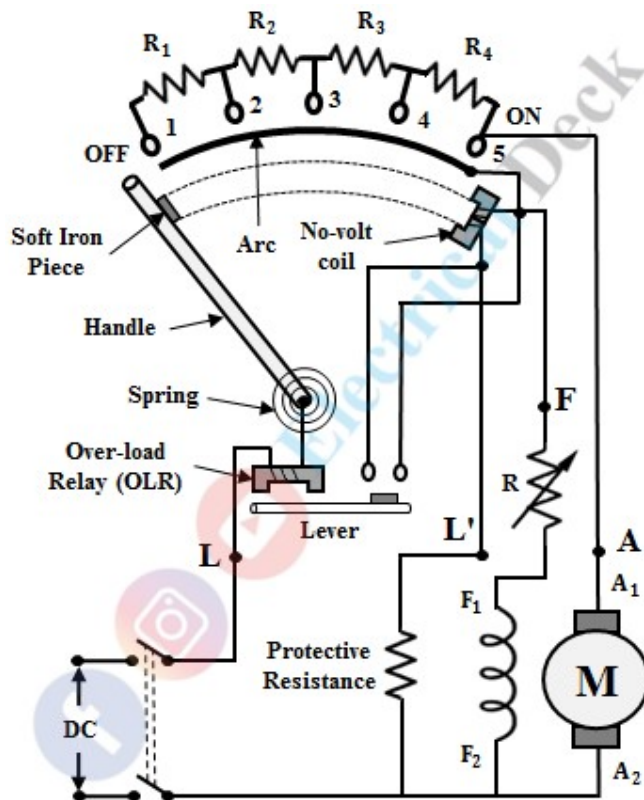
If one transformer breaks down in a star-star connected system of 3 single-phase transformers in a substation, three-phase power cannot be supplied until the defective transformer has been replaced or repaired. To eliminate this undesirable condition, single-phase transformers are generally **connected in Δ - Δ** .

In this case, if one transformer breaks down, it is possible to continue supplying three-phase power with the other two transformers because this arrangement maintains correct voltage and phase relations on the secondary.



However, **with two transformers, the capacity of the bank is reduced to 57.7%** of what it was with all three transformers in service (i.e., complete Δ - Δ circuit)

VIII. A four-point starter protects armature by limiting heavy currents at the time of starting a dc shunt or compound motor. Most of the construction part of a four-point starter is similar to that of a 3-point starter. A little different to that of a 3-point starter, that makes a 4-point starter is by directly connecting NVC across the supply through protective resistance. The connection with this arrangement with a dc shunt motor is shown below.



In the above figure we can see that instead of connecting NVC with the field winding, it is connected separately across the supply that makes a parallel arrangement with the field winding. This creates an additional terminal L' that connects NVC with the supply. This arrangement brings out the starter with some functional difference and four terminals compared to a 3-point starter thus called 'Four-point Starter'. The four terminals of the four-point starter are,

- Terminal-L (line or positive terminal),
- Terminal-A (armature terminal),
- Terminal-F (field terminal),
- Terminal-L' (no-volt coil terminal).

The positive terminal is connected to the over-load release (OLR is used for overload protection as seen in 3-point starter construction). The other end of OLR is connected to the starter handle which is movable over series of resistors terminals (1, 2, 3, 4, 5 as shown in the

above figure). An arc made up of copper or brass is placed below the resistors which supply current to the field winding and no-volt coil. The arc maintains contact with the current carrying handle that moves over it.

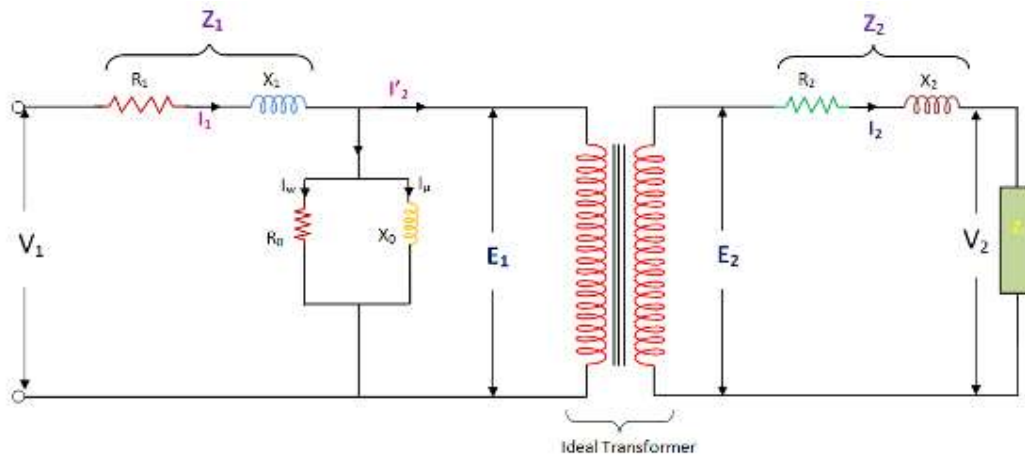
Working of Four-point Starter :

When the motor is connected across the supply by moving the starter handle to the starting resistor terminal. The current from the terminal L flow through the OLR (over-load release) and then to the starter handle. Now the current from the handle is divided into three different paths.

- A part of the supply current flows to the armature (terminal-A) through series of starter resistors,
- A part of the supply current flows through field winding (terminal-F) of the motor, and
- The remaining part of the supply current flows through the NVC (terminal-L').

Here, the point to be noted that, change in field current doesn't affect the electromagnetic pull force on the handle by the NVC coil. Because the NVC circuit (NVC current path) is independent of the field circuit (field current path). Due to this independency, a protective resistance is connected in series with the NVC to regulate current flow, when the field circuit is accidentally kept open while starting the motor. Therefore, the starting current of the motor is regulated and any change in the field excitation doesn't affect the NVC coil thereby holding the starter handle in ON position throughout the operation of motor

IX.The equivalent circuit diagram of transformer is given below:-



****Equivalent Circuit diagram of Transformer****

Where,

R_1 = Primary Winding Resistance.

R_2 = Secondary winding Resistance.

I_0 = No-load current.

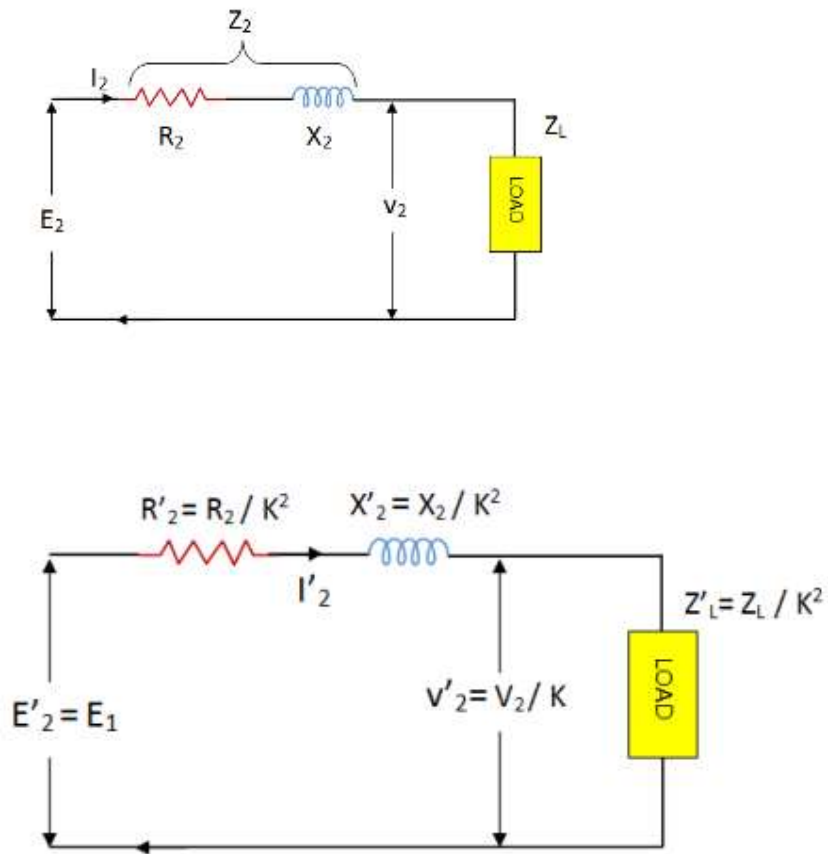
I_μ = Magnetizing Component,

I_w = Working Component,

This I_μ & I_w are connected in parallel across the primary circuit. The value of E_1 (Primary e.m.f) is obtained by subtracting vectorially $I_1 Z_1$ from V_1 . The value of $X_0 = E_1 / I_0$ and $R_0 = E_1 / I_w$. We know that the relation of E_1 and E_2 is $E_2 / E_1 = N_2 / N_1 = K$, (transformation Ratio)

From the equivalent circuit , we can easily calculate the total impedance of to transfer voltage, current, and impedance either to the primary or the secondary.

The secondary circuit is shown in fig-1. and its equivalent primary value is shown in fig- 2,



The total equivalent circuit of the transformer is obtained by adding in the primary impedance as shown in Fig-3 .

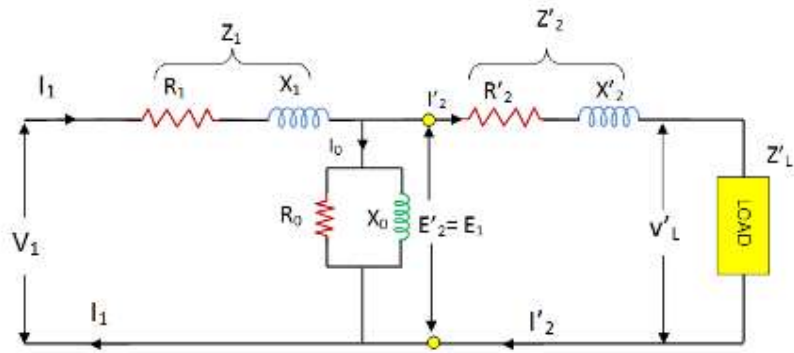


fig-3



And It can be simplified the terminals shown in fig – 4 & further simplify the equivalent circuit is shown in fig.- 5 ,

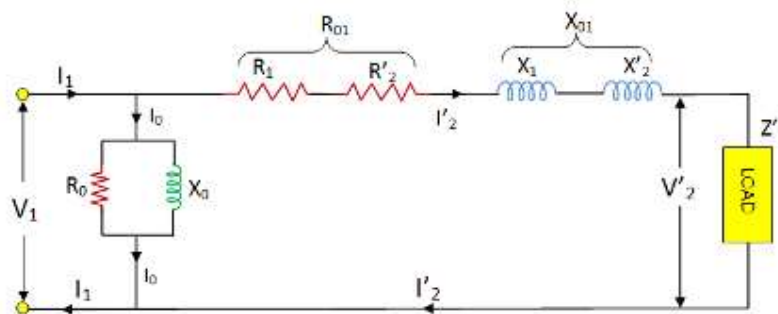
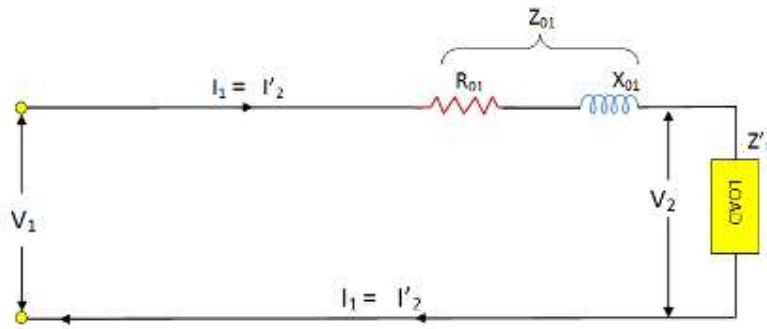


fig-4

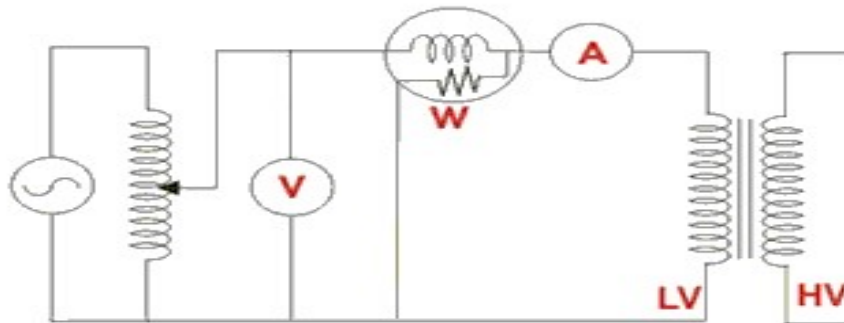


At last, the circuit is simplified by omitting I_0 altogether as shown in fig- 5 .



X. The connection diagram for open circuit test on transformer is shown in the figure. A voltmeter, wattmeter, and an ammeter are connected in LV side of the transformer as shown. The voltage at rated frequency is applied to that LV side with the help of a variac of variable ratio auto transformer.

The HV side of the transformer is kept open. Now with the help of variac, applied voltage gets slowly increased until the voltmeter gives reading equal to the rated voltage of the LV side. After reaching rated LV side voltage, we record all the three instruments reading (Voltmeter, Ammeter and Wattmeter readings)



Open Circuit Test on Transformer

The ammeter reading gives the no load current I_e . As no load current I_e is quite small compared to rated current of the transformer, the voltage drops due to this current that can be taken as negligible.

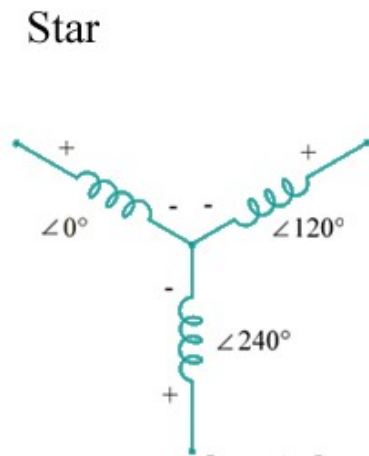
XI. The transformer will give the maximum efficiency when **their copper loss is equal to the iron loss**. Copper loss: A loss in a transformer that takes place in the winding resistance of a transformer is known as the copper loss.

XII. An open delta connection **transformer** uses two single-phase transformers to provide a three-phase supply to the load. An open delta connection system is also called a V-V system. Open delta connection systems are usually only used in emergency conditions, as their efficiency is low when compared to delta-delta (closed delta) systems (which are used during standard operations).

Suppose you have three **single-phase transformers** of 10 kVA each. They are connected in (both primary & secondary sides) in a **delta connection**, then it can be said they are connected in a closed delta system.

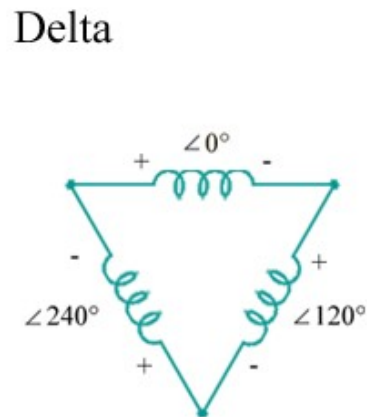
XIII.Voltage Regulation of single-phase transformers is the percentage (or per unit value) change in its secondary terminal voltage compared to its original no-load voltage under varying secondary load conditions. In other words, regulation determines the variation in secondary terminal voltage which occurs inside the transformer as a result of variations in the transformers connected load thereby affecting its performance and efficiency if these losses are high and the secondary voltage becomes too low.

XIV.Three phase transformer connections In three phase system, the three phases can be connected in either star or delta configuration. In case you are not familiar with those configurations, study the following image which explains star and delta configuration. In any of these configurations, there will be a phase difference of 120° between any two phases.



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Center point must tie either all '-' or all '+' ends together.



The winding polarities must stack together in a complementary manner (+ to -)

Three Phase Transformer Connections

Windings of a **three phase transformer** can be connected in various configurations as (i) star-star, (ii) delta-delta, (iii) star-delta, (iv) delta-star, (v) open delta and (vi) Scott connection. These configurations are explained below.

Star-Star (Y-Y)

- Star-star connection is generally used for small, high-voltage transformers. Because of star connection, number of required turns/phase is reduced (as phase voltage in star connection is $1/\sqrt{3}$ times of line voltage only). Thus, the amount of insulation required is also reduced.
- The ratio of line voltages on the primary side and the secondary side is equal to the **transformation ratio** of the transformers.
- Line voltages on both sides are in phase with each other.
- This connection can be used only if the connected load is balanced.

Delta-Delta (Δ - Δ)

- This connection is generally used for large, low-voltage transformers. Number of required phase/turns is relatively greater than that for star-star connection.
- The ratio of line voltages on the primary and the secondary side is equal to the transformation ratio of the transformers.
- This connection can be used even for unbalanced loading.
- Another advantage of this type of connection is that even if one transformer is disabled, system can continue to operate in open delta connection but with reduced available capacity.

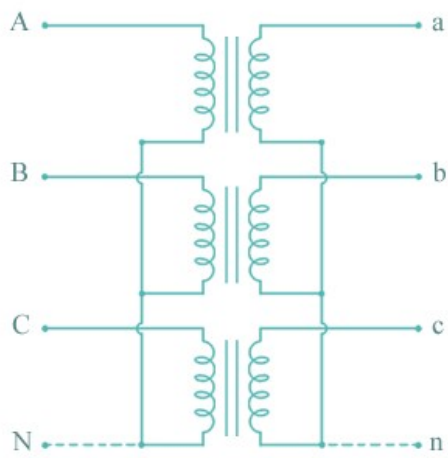
Star-Delta OR Wye-Delta (Y- Δ)

- The primary winding is star star (Y) connected with grounded neutral and the secondary winding is delta connected.
- This connection is mainly used in step down transformer at the substation end of the transmission line.
- The ratio of secondary to primary line voltage is $1/\sqrt{3}$ times the transformation ratio.
- There is 30° shift between the primary and secondary line voltages.

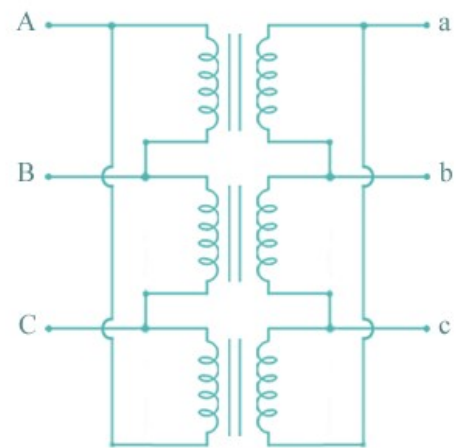
Delta-Star OR Delta-Wye (Δ -Y)

- The primary winding is connected in delta and the secondary winding is connected in star with neutral grounded. Thus it can be used to provide 3-phase 4-wire service.
- This type of connection is mainly used in step-up transformer at the beginning of transmission line.
- The ratio of secondary to primary line voltage is $\sqrt{3}$ times the transformation ratio.
- There is 30° shift between the primary and secondary line voltages.

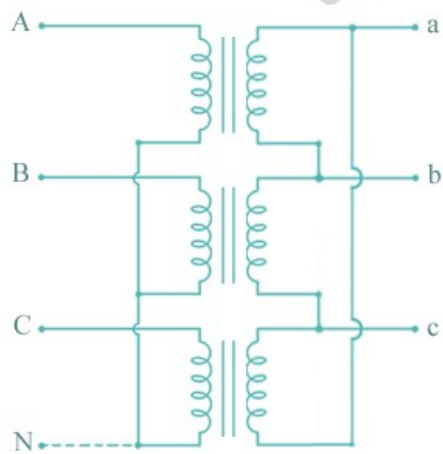
Above transformer connection configurations are shown in the following figure.



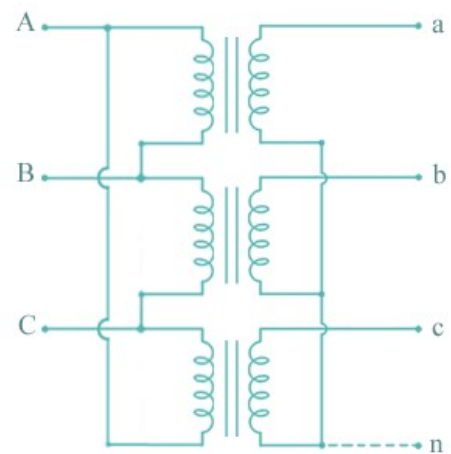
Y - Y



Δ - Δ



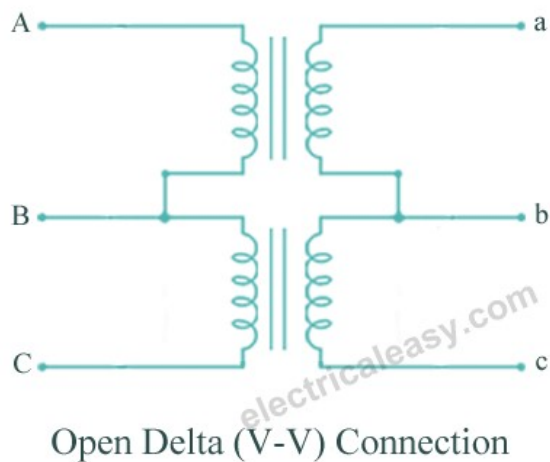
Y - Δ



Δ - Y

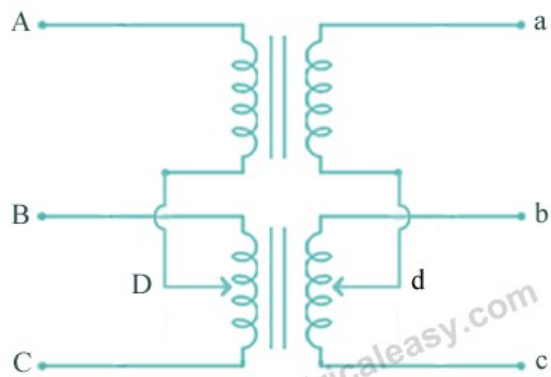
Open Delta (V-V) Connection

Two transformers are used and primary and secondary connections are made as shown in the figure below. Open delta connection can be used when one of the transformers in Δ - Δ bank is disabled and the service is to be continued until the faulty transformer is repaired or replaced. It can also be used for small three phase loads where installation of full three transformer bank is unnecessary. The total load carrying capacity of open delta connection is 57.7% than that would be for delta-delta connection.



Scott (T-T) Connection

Two transformers are used in this type of connection. One of the transformers has centre taps on both primary and secondary windings (which is called as main transformer). The other transformer is called as teaser transformer. Scott connection can also be used for three phase to two phase conversion. The connection is made as shown in the figure below.



Scott (T-T) Connection

